MIL-A-25915A (USAF)
30 APRIL 1980
SUPERSEDING
MIL-A-25915 (USAF)
8 November 1957

MILITARY SPECIFICATION

ACCELEROMETER, AIRCRAFT, PILOT'S WARNING, TYPE MA-2

This specification is approved for use by code 99, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of 1-7/8 inch dial, -2 to +8 g range accelerometer, designated Type MA-2.

2. APPLICABLE DOCUMENTS

*2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation-packaging, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-S-6872	Soldering Process, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6610

Source: https://assist.dla.mil -- Downloaded: 202610-20T07:30Z
Check the source to verify that this is the current version before use.

MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-L-25142	Luminescent Material, Fluorescent
MIL-C-83488	Coating, Aluminum, Ion Vapor Deposited

STANDARDS

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking For Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-889	Dissimilar Metals
MS 33558	Numeral and Letter, Aircraft Instrument Dial, Standard Form of
MS 33585	Pointers, Dial, Standard Dimensions of Aircraft Instrument
MS 33638	Case, Instrument, Flange Mounted, Aircraft

(Copies of specifications, standards, drawing and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 First article. When specified, the contractor shall furnish sample units for first article inspection and approval (see 4.2 and 6.3).

*3.2 Selection of specifications and standards. Specification and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial Parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins having suitable properties may be used provided:

Source: https://assist.dla.mil -- Downloaded: 202610-20T07:30Z
Check the source to verify that this is the current version before use.

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.2.2 Standard parts. With the exception in 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 MATERIALS

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts except where magnetic materials are essential.

3.3.3 Nonferrous materials. Nonferrous materials shall be used for all parts of the accelerometer except where ferrous materials are essential.

3.3.4 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

*3.3.5 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.6 Corrosive fumes. The materials as installed in the accelerometer shall not liberate deleterious fumes.

3.3.7 Protective treatment. When materials are used in the construction of the accelerometer that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

Source: https://assist.dla.mil -- Downloaded: 202610-20T07:30Z
Check the source to verify that this is the current version before use.

3.4 Design and construction. The accelerometer shall be designed to indicate aircraft accelerations in maneuvers and rough air and shall conform to figure 1. The accelerometer shall be designed to meet the response characteristics specified in section 4. The construction of the accelerometer shall be reasonably simple to permit overhaul or repair without the use of special tools or fixtures.

3.4.1 Indication and range. The accelerometer shall indicate vertical accelerations when the plane of the mounting lugs is vertical and the +1 graduation is at 270^{0}. When in this position, normal gravity shall be indicated at +1 g. The accelerometer shall be designed to indicate accelerations in the range -2 to +8 g units.

*3.5 Performance. The accelerometer shall be capable of meeting the test requirements specified in section 4 when subjected to the following conditions:

a. Temperature. - Temperature ranging from $+71^{0}$C to -62^{0}C.

b. Humidity. - Relative humidity up to 100 percent.

c. Altitude. - Pressures ranging from 30 inches Hg down to 1.31 inches Hg approximately 70,000 feet altitude.

d. Fungus. - Fungus growth as encountered in tropical climates.

e. Salt spray. - Exposure to salt sea atmosphere for a period of 50 hours.

f. Thermal shock. - Thermal shock with alternate immersion in water at temperatures of $+85^{0}$C and $+5^{0}$C for 8 cycles.

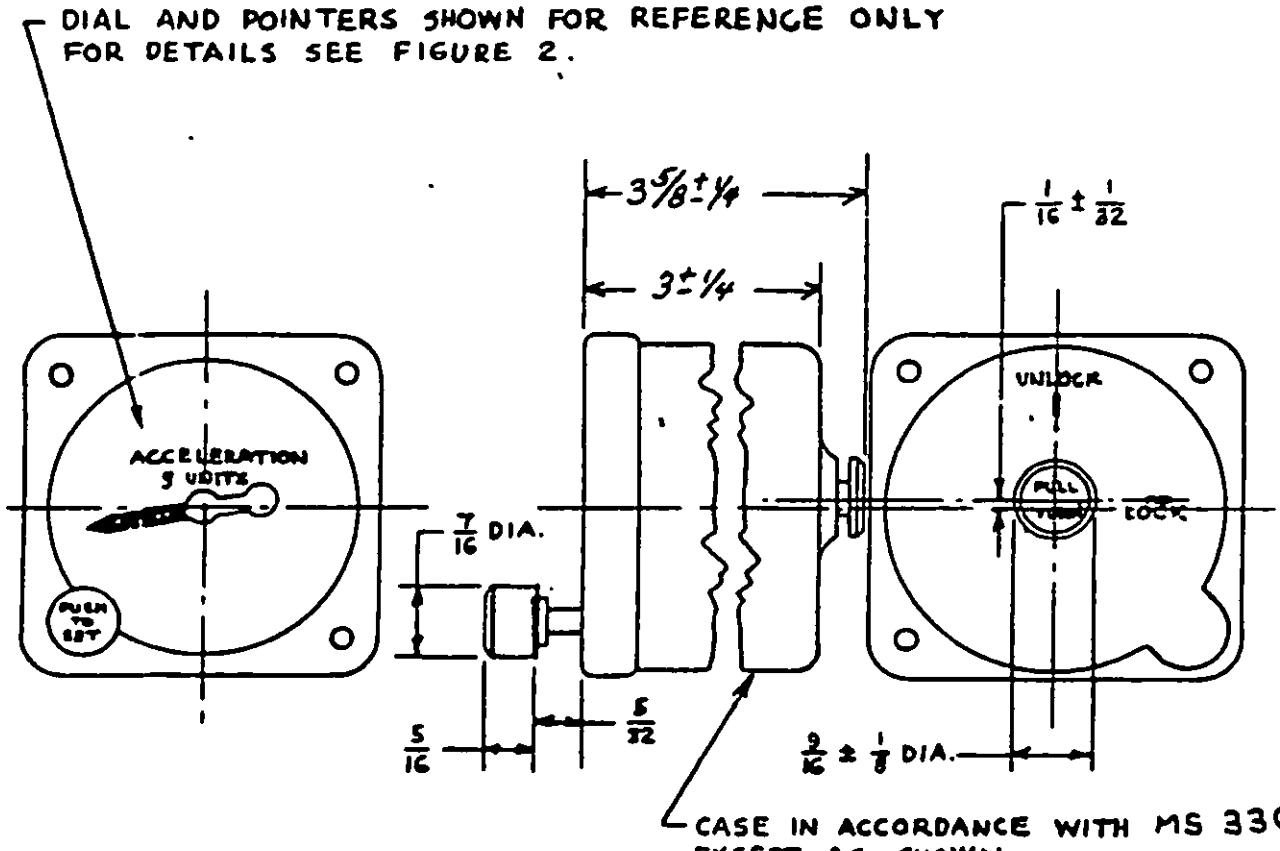
g. Sealing. - Immersion in water with pressure applied and without leakage.

h. Acceleration and shock. - Acceleration and shock forces as anticipated.

i. Vibration error. - When vibrated at an amplitude of 0.018 to 0.020 inch and at a frequency variation between 500 and 3,000 cpm for a total of 3 hours the main pointer shall not exceed the 0.2 g to 0.4 g range and the auxiliary pointers shall not change more than 0.2 g.

j. Vibration endurance. - The dynamic scale error shall not exceed 0.2 g when subjected to the vibration endurance test.

Source: https://assist.dla.mil -- Downloaded: 202640-20T07:30Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED, TOLERANCES: FRACTIONS ± 1/64

FIGURE 1. CASE

Source: https://assist.dla.mil -- Downloaded: 202610-20T07:30Z
Check the source to verify that this is the current version before use.

k. Static scale error and friction. - Static scale error and friction shall not exceed ±0.1 g and ±0.2 g, respectively when tested as specified.

l. Dynamic scale error. - Dynamic scale error shall not exceed ±0.2 g at any specified test point and ±0.1 g at 70 percent of the test points.

m. Magnetic effect. - With the accelerometer revolved about a magnetic compass with a magnetic field intensity of 0.17 to 0.19 oersted, the compass shall not deflect more than 1^{0}.

n. Mounting lugs. - The mounting lugs shall withstand a load of 175 pounds without fracture.

o. Response. - Response characteristics shall be within the limits specified in 4.5.5, when the accelerometer is vibrated at frequencies from 4 to 20 cps at maximum g forces for each test point.

3.5.1 The accelerometer shall be so constructed as to withstand the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service, without failure.

3.5.1.1 Pivots, bearings, and gears shall neither bind nor shake and shall be as near frictionless as practicable.

*3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-STD-100 shall govern changes in the manufacturer's part numbers.

3.7 Dial. The dial shall conform to figure 2 and shall be fastened securely to the case or upon the frame of the mechanism by not less than two screws in such a manner that it will not loosen or turn when the instrument is vibrated.

3.7.1 Distance from dial to cover glass. Distance between dial and cover glass shall not be greater than 0.250 inch.

3.7.2 Numerals and letters. The style and proportion of numerals and letters, except the letter "g", placed on the dial of the accelerometer shall conform to MS33558. Numerals shall distinctly indicate the graduation to which each applies.

Source: https://assist.dla.mil -- Downloaded: 202610-20T07:30Z
Check the source to verify that this is the currentversion before use.

3.8 Pointers. The accelerometer shall have three pointers. One main pointer for indicating instantaneous acceleration and two auxiliary pointers; one for indicating maximum positive acceleration; and one for indicating maximum negative acceleration. The pointers shall conform to figure 2. They shall be light and sufficiently rigid to prevent oscillation under vibration. They shall be firmly attached to the mechanism, yet shall be readily adjustable.

3.8.1 Ratchet mechanism. Each auxiliary pointer shall be provided with a ratchet device, in order that indication of maximum plus and minus deflections of the main pointer will be maintained until the auxiliary pointers are reset to obtain a new reading.

3.8.2 Auxiliary pointer return mechanism. A mechanism shall be provided for resetting the auxiliary pointers into juxtaposition with the main pointer. A knob, located as shown on figure 1, shall be provided for operating the return mechanism. Stops shall be provided in order that the auxiliary pointers when reset shall not indicate a lower or higher reading than that indicated by the main pointer. Stops shall be a part of the mechanism connecting the pointers and shall not directly engage them. Operation of the return mechanism shall require 2-1/2 to 8 pounds of force on the knob.

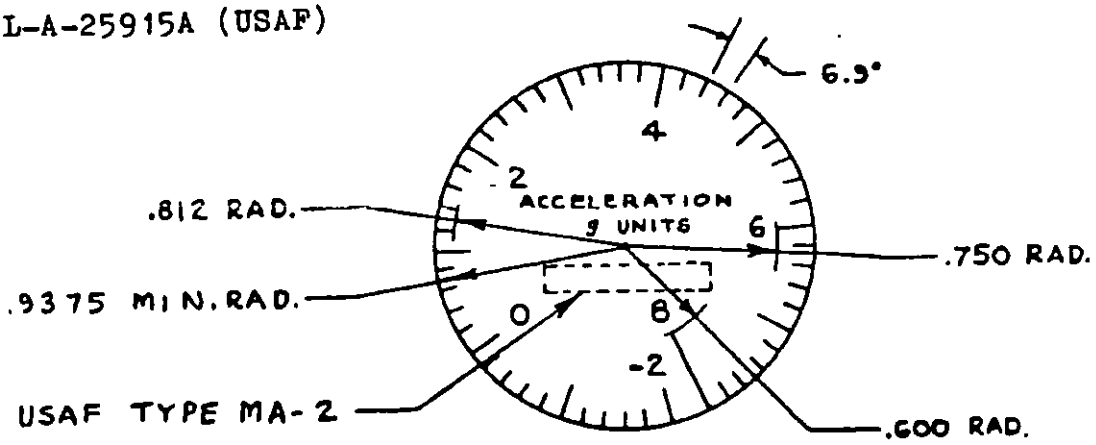
3.8.3 Main pointer mechanism. The main pointer mechanism shall not engage the mechanism of the auxiliary pointers when the imposed acceleration is less than the acceleration indicated by the auxiliary pointers.

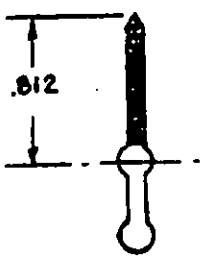
3.9 ACCELERATION-RESPONSIVE ELEMENT

3.9.1 Stops for acceleration-responsive element. Mechanical stops shall be provided for arresting the motion of the acceleration-responsive element when the accelerometer is subjected to accelerations beyond the limits specified. One stop shall engage the element at an acceleration of not less than -2.1 g and more than -2.2 g, and the other at an acceleration of not less than +8.1 g and not more than +8.2 g.

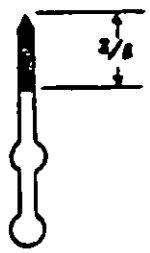
3.9.2 Locking device. A device for manually locking the position of the acceleration-responsive element shall be provided, if the design is such as to require it in order to protect the mechanism of the accelerometer from possible damage during shipment. If provided, this device shall be controlled by means of a knurled knob located in the back of the case as shown on figure 1.

Source: https://assist.dla.mil -- Downloaded: 202610-20T07:30Z
Check the source to verify that this is the current version before use.





MAIN POINTER
MS33586-1



AUXILIARY POINTER
MS33585-1
EXCEPT AS SHOWN

DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES: FRACTIONS ± 1/64
DECIMALS ±.010
ANGLES ± 0° 10'

FIGURE 2. DIAL AND POINTERS

Source: https://assist.dla.mil -- Downloaded: 202610-20T07:30Z
Check the source to verify that this is the current version before use.

3.9.3 Adjustment. The mechanism shall be provided with means to adjust or correct the indications. This means of adjustment shall be reasonably simple to manipulate and of such a nature that adjustment can be accomplished with tools ordinarily possessed by an instrument repairman.

3.10 Case. The case shall conform to figure 1 and shall be finished in durable dull black. The case shall be constructed of a suitable material applicable to hermetic sealing, uniform in texture, and which shall provide a smooth surface.

3.11 Hermetic sealing. The indicator shall be hermetically sealed. A tear strip shall be provided to facilitate removal and resealing of the mechanism. There shall also be a filling tube of a malleable variety of metal which shall be capable of being formed into a slot provided on the rear of the case so as to be flush with the rear surface when not in use.

3.11.1 Filling medium. In the hermetically sealed case the filling medium shall be of at least 98 percent purity, free of dust particles, and containing not more than 0.006 milligrams of water vapor per liter (dew point -65°C) at the filling pressure. The filling medium shall be either 100 percent helium or a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be one atmosphere. Where practicable the 100 percent helium filling medium shall be utilized.

3.12 Marking. All markings shall be durable to withstand the usage encountered in service. Numerals shall distinctly indicate the graduation to which each applies.

3.12.1 Fluorescent-luminescent material. The following markings shall be finished in fluorescent-luminescent material in accordance with MIL-L-25142, type I or III, as applicable.

	1/64 3/16	Inch +0.005 MS33558
Numerals		
Graduations corresponding to numerals	See Figure 2	0.031
Lettering "g UNITS" and "ACCELERATION:	1/16	MS33558
One "g" Graduation		0.062
Shaded part of pointers		
Knob markings		
Small graduations	See Figure 2	0.015

Source: https://assist.dla.mil -- Downloaded: 202610-20T07:30Z
Check the source to verify that this is the current version before use.

3.12.2 Durable dull black. The following markings shall be durably and legibly marked on the dial. These markings, the background of the dial, and all other markings not otherwise specified shall be finished in durable dull black.

	Height or Length Inch - 1/64
Lettering "USAF" and "TYPE MA-2"	1/16

3.12.3 Visibility of dial. The pointers, numerals, at least 1/16 inch of the shortest graduations, and other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 30^{0} with a perpendicular to the dial and whose small diameter is the aperture of the instrument case.

3.13 Soldering. Soldering shall be accomplished in accordance with MIL-S-6872.

*3.14 Lubrication. The accelerometer shall be lubricated in accordance with MIL-STD-838.

3.15 Weight. The weight of each accelerometer shall not exceed 1.50 pounds.

3.16 FINISHES AND PROTECTIVE COATINGS

3.16.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625 except as follows:

a. Dials, small holes, and case inserts need not be anodized.

b. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

c. Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

d. Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

e. When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

Source: https://assist.dla.mil -- Downloaded: 20260-20T07:30Z
Check the source to verify that this is the current version before use.

*3.16.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used.

3.18. Workmanship. The accelerometer, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, and riveting; alignment of parts and tightness of assembly screws, and bolts, etc.

3.18.1 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciable tightened further without damage or injury to the screw or bolt or threads.

3.18.2 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.18.3 Gears. Gear assemblies shall be properly aligned and meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

3.18.4 Cleaning. The accelerometer shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

3.18.5 Dimensions. Dimensions and tolerances not specified, shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability operation, or performance of the accerlerometer they shall be held or limited accordingly.

3.19 Envelope. An envelope, containing the mounting screws, shall be furnished with each indicator. The following information shall be printed on the face of the envelope:

Source: https://assist.dla.mil -- Downloaded: 202610-20T07:30Z
Check the source to verify that this is the current version before use.

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Classification of tests. The inspection and testing of the accelerometer shall be classified as follows:

a. Acceptance tests. Acceptance tests are those tests accomplished on accelerometers submitted for acceptance under contract.

b. First article. First article tests are those tests done on items representative of production equipment to check for conformance to specification requirements (see 4.6).

4.3 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests.

b. Sampling plans A and B tests.

4.3.1 Individual tests. Each accelerometer shall be subjected to the following tests as described under 4.5 TEST METHODS:

a. Examination of product.

b. Static scale error and friction

c. Dynamic scale error.

d. Sealing.

e. Response characteristics (4.5.5.1)

f. Return mechanism operation.

Source: https://assist.dla.mil -- Downloaded: 202610-20T07:30Z
Check the source to verify that this is the current version before use.

4.3.2 SAMPLING PLANS AND TESTS

4.3.2.1 Sampling plan A. One accelerometer selected at random from each 100 or less produced on the contract or order shall be subjected to the following tests as described under 4.5 TEST METHODS:

a. Individual tests.

b. Low temperature operation

c. High temperature operation.

d. Magnetic effect.

e. Vibration error.

f. Response characteristics (4.5.5)

4.3.2.2 Sampling plan B. Unless otherwise specified, three accelerometers selected at random from the first 15 items of the contract or order shall be subjected to the following tests as described under 4.5 TEST METHODS:

a. Sampling plan A tests.

b. Vibration endurance.

c. Shock.

d. Mounting lugs.

e. Low temperature - high altitude.

f. High temperature exposure.

g. Thermal shock.

h. Humidity.

i. Fungus.

j. Salt spray.

4.3.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:30Z
Check the source to verify that this is the current version before use.

4.3.2.3.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.3.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods or correcting them.

4.4 TEST CONDITIONS

4.4.1 Standard atmospheric conditions. When ever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

*4.4.2 Tapping. Unless otherwise specified, the accelerometer shall be tested in the normal operating position and shall be gently tapped or vibrated before a test reading is taken. Tapping shall be applied in a Longitudinal direction with respect to the dial of the instrument.

4.4.3 Vibration stand. A vibration stand shall be used which will vibrate at any desired frequency between 500 and 3000 cpm and shall subject the accelerometer to vibration such that a point on the case will describe in a plane inclined 45° to the horizontal plane, a circle, the diameter of which is equal to the amplitude specified herein.

4.4.4 Calibration procedure. The contractor shall furnish a drawing, with the qualification sample, of the mechanism and case showing clearly the vertical distance of the center of mass of the actuating mechanism from the center of the instrument dial for each of the required test points. This drawing shall also include a table for the revolutions per minute required for each test point computed on the basis of a rotating arm whose radius to the center of the instrument dial is 22 inches. The revolutions per minute required for each test point shall be computed by the formula:

$$N = 9.5493 \sqrt{\frac{32.2 \text{ "g"}}{r}}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:30Z
Check the source to verify that this is the current version before use.

Where N = RPM of rotating
"g" = Test point
r = Radius of arm in feet, corrected for mass position

4.4.4.1 Test points. "N" shall be computed for all points from -2 g to +8 g in increments of 1 g.

4.4.5 No adjustments or repairs of any type shall be performed the qualification test samples after start of qualification test

4.5 TEST METHODS

4.5.1 Examination on product. Each accelerometer shall be examined externally to determine conformance with this specification with respect to material, workmanship, marking, and envelope.

*4.5.2 Static scale error and friction. The position error shall be observed with accelerometer held in each of the following positions:

a. Plane of dial vertical with +1 g graduation at 270^{0}.

b. Plane of dial horizontal.

c. Plane of dial vertical with +1 g graduation at 90^{0}.

In each position, the main-pointer reading obtained shall be due entirely to gravity and not to motion in change of position of instrument. The static scale error shall not exceed ±0.1 g in any of the three positions tested. The accelerometer shall also be tested for friction at the three positions mentioned above. The amount of tapping or vibration applied shall be sufficient to remove all friction from the accelerometer. In going from one position to the next, one of the memory pointers shall be engaged. After the desired position is reached, two readings shall be taken, the first before vibration and the second after vibration. The difference of two such readings shall be recorded as friction and shall not exceed 0.2 g with the plane of the dial vertical and 0.3g with the plane of the dial horizontal test points.

4.5.3 Dynamic scale error. The accelerometer shall be mounted on a suitable centrifuge and the scale errors determined for accelerations of -2, -1, +1, +2, +3, +4, +5, +6, +7, and +8 g and at any other points deemed necessary to determine accuracy. The dynamic scale error shall not exceed ±0.2 g at any test point. The error shall not exceed ±0.1 g for 70 percent of the test points.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:30Z
Check the source to verify that this is the current version before use.

4.5.4 Sealing. The accelerometer case shall be tested for leaks by means of a mass spectrometer type of helium leak detector and the differential pressure existing when the test is conducted shall be the same as that to which the instrument is filled, approximately one atmosphere. The leak rate shall be based on one atmosphere differential pressure and pure helium as the filling medium. Under these conditions the leak rate shall not exceed 0.10 micron cubic foot per hour. Equivalent helium leak rates for the same hole size at other differential pressures and helium-nitrogen mixtures, or both, shall be calculated by the manufacturer.

*4.5.5 Response characteristics (Sampling plan A). The accelerometer shall be subjected to vibration along the normal axis, which is perpendicular to the longitudinal and lateral axes. The test apparatus used shall be as specified in 4.5.5.2. The accelerometer shall be tested at each of a minimum of 17 diferent frequencies. These tests shall be approximately equally spaced through the frequency range from 4 to 20 cps. The amplitude of vibration at each test point shall be such that the maximum g forces practicable for that test point shall be obtained. The indicated g shall be recorded and a plot made for each test point of indicated g/ applied g vs frequency. The points so plotted shall fall within the shaded area of graph on figure 3. These graphs shall be a part of the qualification test data.

4.5.5.1 Response characteristics (Individual tests). This test shall be identical to the test specified in 4.5.5 except that the accelerometer shall be tested at the following frequencies only: 4, 6, 8, 12, 16, and 20 cps.

4.5.5.2 Test apparatus. The equipment shall impose linear sinusoidal motion to the accelerometer and shall be capable of procuring the frequencies and double amplitude required for each test point. The acceleration vector of any harmonic measured on the test table shall not exceed 10 percent of the fundamental acceleration vector.

4.5.6 Return mechanism operation. The accelerometer shall be placed in operating position with the pointer deflected to the maximum indication, the reset knob shall then be operated to return the pointer to zero. The force applied to the reset knob shall be between 2-1/2 and 8 pounds.

4.5.7 Low temperature operation. The accelerometer shall be subjected to the low temperature test, procedure I of MIL-E-5272 for a period of 4 hours. Upon completion of the test period and while at the test temperature the accelerometer shall be subjected to and meet the individual tests.

Source: https://assist.dla.mil -- Downloaded: 202610-20T07:30Z
Check the source to verify that this is the current version before use.

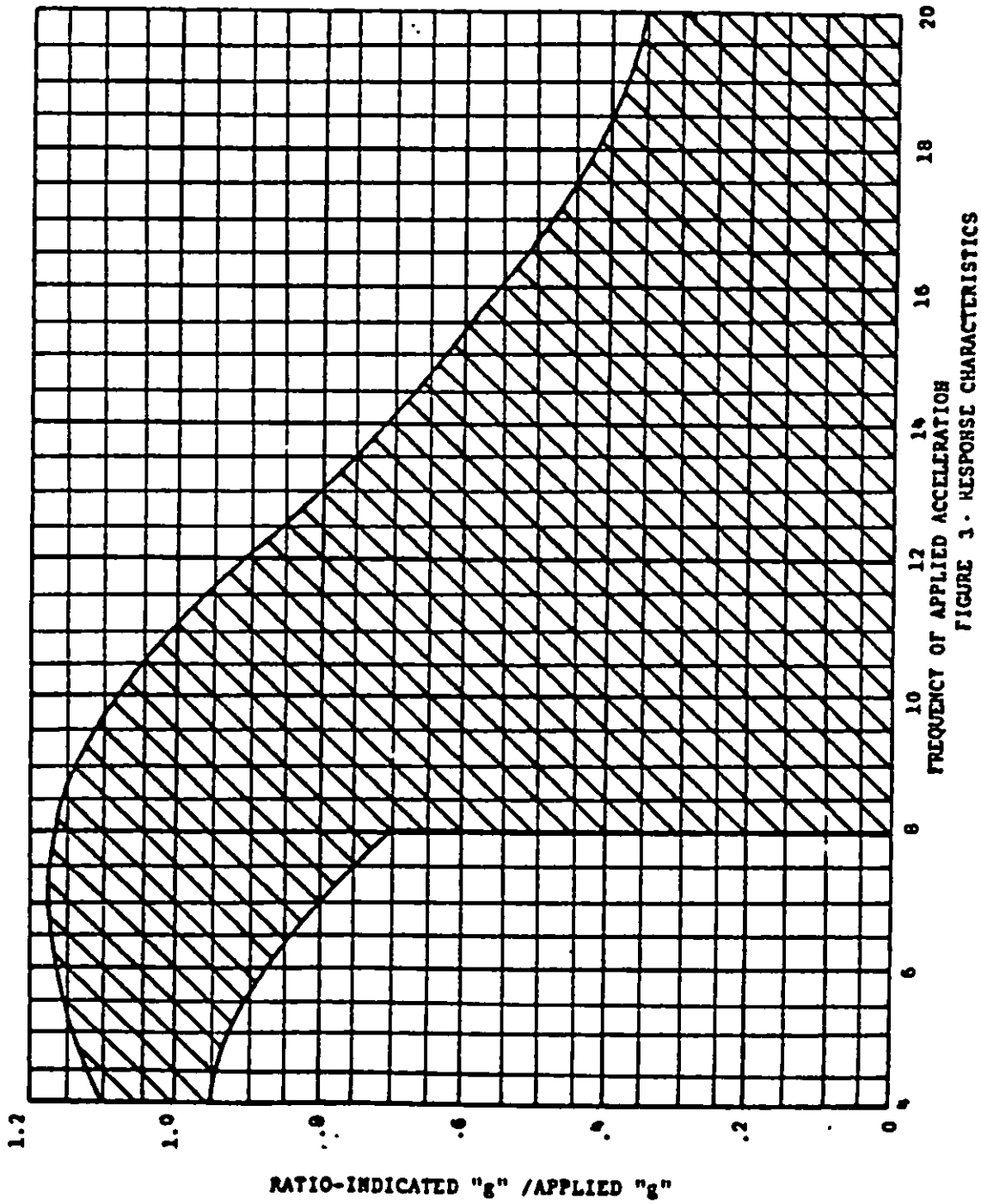


FIGURE 3. RESPONSE CHARACTERISTICS

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:30Z
Check the source to verify that this is the current version before use.

4.5.8 High temperature operation. The accelerometer shall be subjected to the high temperature test, procedure II of MIL-E-5272 for a period of 4 hours. Upon completion of the test period and while at the test temperature the accelerometer shall be subject to and meet the individual tests.

4.5.9 Magnetic effect. The accelerometer shall be revolved about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from the bar magnet, and in a vertical plane which is a perpendicular bisector of the North-South axis of the bar magnet. Starting directly under the compass, the indicator shall be held in positions 0^0, 45^0, 90^0, 135^0, 180^0, 225^0, 270^0, and 315^0 from the initial position. At each of these positions, the indicator shall be rotated on its horizontal axis until it is in its normal operating position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 1^0.

4.5.10 Vibration error. The accelerometer shall be mounted on a suitable vibration stand with the dial in a vertical plane and with the graduation +1 at 270^0. The auxiliary pointers shall be displaced approximately 1 g from the main pointer and the readings noted. The reset mechanism shall not be operated until the completion of the tests. For a total of 3 hours the accelerometer shall be vibrated at the specified frequencies at an amplitude between 0.018 and 0.020 inch, and the pointer oscillation shall be noted. The frequency shall be varied from 500 to 3000 cycles per minute in a period of not more than 60 minutes. The natural frequency of the seismic system shall be noted during the frequency variation between 500 and 3000 cpm. For the remainder of the 3 hour test period, the accelerometer shall be vibrated at a frequency of 1,800 cpm. At the end of the test period, the indication of the auxiliary pointer shall be read before operating the reset mechanism. At frequencies below 1,200 cpm the oscillation of the main pointer shall not exceed 0.2 g and at frequencies over 1,200 cpm, the oscillation shall not exceed 0.3 g. After vibration, the indications of the auxiliary pointers shall not be changed by more than 0.1 g. The accelerometer shall not be damaged as a result of this test.

4.5.11 Vibration endurance. The accelerometer shall be subjected to the vibration test, procedure V of MIL-E-5272 for a period of 50 hours. Upon completion of the test period the accelerometer shall be subjected to the test specified in 4.5.3. The dynamic scale error shall not exceed 0.2 g.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:30Z
Check the source to verify that this is the current version before use.

4.5.12 Shock: The accelerometer shall be installed in the normal operating position in a free-fall apparatus as described in 4.5.11.1. If provided with a locking device, the accelerometer mechanism shall be locked. The accelerometer shall be allowed to fall from a height of 8 inches, then examined for damage and subjected to the test specified in 4.5.3. There shall be no damage to the accelerometer and the steady state scale error shall not exceed 0.2 g.

4.5.12.1 Apparatus. A rigidly constructed device provided with a panel mounting for installing the accelerometer in the specified position, and vertical guides to allow free-fall from the specified height shall be used. A bumper on the free-fall element shall arrest its fall by striking the center of a rounded projection on a metal block. The bumper shall be spring steel, 1/4 by 1-1/2 inches in cross section with an unsupported length of 5 inches between points of attachment to the free-fall element. The bumper shall contact the metal block at the midpoint of a broad side. The center of gravity of the instrument under test and the contact point of the bumper shall be aligned vertically within 1/8 inch. The weight of the free-fall element shall be approximately 12 pounds. The metal block shall weigh not less than 50 pounds.

4.5.13 Mounting lugs. The accelerometer case with mechanism and cover glass removed shall be mounted face downward on the movable head of a suitable testing machine with the diameter of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug and attached to a pull strap in the stationary head of the machine. A load of 175 pounds shall be applied along the axis of the mounting hole and toward the front of the case for a period of 1 minute. There shall be no fracture.

4.5.14 Low temperature - high altitude. The accelerometer shall be subjected to the low temperature test, procedure VI of MIL-E-5272, and an altitude of 70,000 feet (1.31 inches Hg). There shall be no damage to the accelerometer as a result of this test.

4.5.15 High temperature exposure. The accelerometer shall be subjected to the high temperature test, procedure II of MIL-E-5272. No damage to the accelerometer shall result from this test.

Source: https://assist.dla.mil -- Downloaded: 202640-20T07:30Z
Check the source to verify that this is the current version before use.

4.5.16 Thermal shock cycling. The accelerometer shall be alternately subjected to a total of 8 cycles of immersion in tap water maintained at $85^{0} \pm 2^{0}C$ and $5^{0} \pm 2^{0}C$. the length of time for each immersion shall be 30 minutes; not more than 5 seconds shall elapse between immersions. No evidence of moisture penetration or damage to the inclosure shall result from this test. Following this test, the accelerometer shall be subjected to and meet the requirements for the test specified in 4.5.4.

4.5.17 Humidity. The accelerometer shall be subjected to the himidity test in accordance with procedure I of MIL-E-5272.

4.5.18 Fungus. The accelerometer shall be subjected to the fungus resistance test in accordance with procedure I of MIL-E-5272.

4.5.19 Salt spray. The accelerometer shall be subjected to the salt spray test in accordance with procedure I of MIL-E-5272 for a period of 50 hours. Upon completion of the 50 hour test period, there shall be no evidence of external corrosion or discoloration as a result of this test.

*4.6 FIRST ARTICLE TESTING

*4.6.1 First article test samples. The first article test samples shall consist of three accelerometers.

*4.6.2 First article tests. First article tests shall consist of the tests specified under 4.5 TEST METHODS.

*4.7 Packaging Inspection.

*4.7.1 Quality conformance. The inspection of the preservation-packaging and interior package marking shall be in accordance with Group A and B quality conformance inspection requirements, Section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be inaccordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

*4.7.2 First article. When the unit container is capable of serving as the shipping container First Article Inspection, and rough handling tests as outlined in Section 4 of MIL-P-116, shall be accomplished followed by a functional test of the unit to insure freedom from operational malfunction.

Source: https://assist.dla.mil -- Downloaded: 202640-20T07:30Z
Check the source to verify that this is the currentversion before use.

5. PACKAGING.

5.1 Preservation-packaging. Preservation-packaging shall be Level A or C as specified.

5.1.1 Level A.

5.1.1.1 Cleaning. Unit shall be cleaned in accordance with Process C-1 of MIL-P-116.

5.1.1.2 Drying. Immediately after cleaning, the unit shall be dried following any one or combination of the drying procedures of MIL-P-116. The drying procedures employed shall not be injurious to the unit.

5.1.1.3 Preservaton application. Not required.

5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each unit shall be packaged in quantity unit packs of one each in accordance with Method IC-3 of MIL-P-116. Cushioning material between bag and container shall be of a type, density and thickness that will protect the unit from impact shock damage when tested as specified in paragraph 4.7 of this specification.

5.1.2 Level C. Units shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

*5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

*5.2.1 Level A. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636 weather-resistant. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, specification PPP-B-601 plywood box, shall be used for the shipping container.

*5.2.2 Level B. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636, class domestic. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, the shipping containers shall conform to specification PPP-B-636 class weather-resistant.

Source: https://assist.dla.mil -- Downloaded: 20260-20T07:30Z
Check the source to verify that this is the current version before use.

*5.2.3 Level C. Packaged units shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to applicable carrier rules and regulations.

*5.3 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The Type MA-2 accelerometer covered by this specification is intended for use on aircraft to indicate vertical acceleration in maneuvers and in rough air.

6.2 DEFINITIONS.

6.2.1 Tapping. The phrase "TAPPED LIGHTLY" is defined as meaning the application of a light vibration sufficient to overcome that amount of residual friction normally associated with good quality aircraft instruments.

6.2.2 Hermetic seal. A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials, or both. A hermetic seal is not intended to include seals accomplished by gaskets.

6.2.3 "g" unit. The "g" unit is defined as an acceleration of 32.2 feet per second, per second.

6.2.4 Amplitude. Whenever the work "AMPLITUDE" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.3 Ordering data. Procurement documents should specify the following:

a. When a first article is required for inspection and approval (see 3.1 and 4.2).

b. Level of packaging and packing desired.

c. Title, number, and date of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:30Z
Check the source to verify that this is the current version before use.

6.4 International standardization agreements. Certain provisions (see figure 1 and 2) of this specification are the subject of international standardization agreement ASCC AIR STD 10/4. When amendment, revision, or cancellation of this specification is proposed which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

*6.5 Reclaimed material. The use of reclaimed materials shall be encouraged to the maximum extent possible.

*6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 99

Project number:
6610-F312

Source: https://assist.dla.mil -- Downloaded: 202610-20T07:30Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective user the document are invited to submit comments to the government. Fold on lines on reverse side, staple in c and send to preparing activity. Attach any pertinent data which may be of use in improving this document there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating t the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specificati requirements on current contracts. Comments submitted on this form do not constitute or imply authoriz to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-A-25915A (USAF)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCU
USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELO

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY (Printed or typed name and address — Optional)	TELEPHONE NO.
	DATE

DD FORM 1 OCT 76 1426 PREVIOUS EDITION WILL BE USED.

Source: https://assist.dla.mil -- Downloaded: 202610-20T07:30Z
Check the source to verify that this is the current version before use.